

6.2 Standard Form Worksheet

MPM1D

1. Express each equation in the form $y = mx + b$

a) $x + y - 3 = 0$

b) $2x + 3y + 6 = 0$

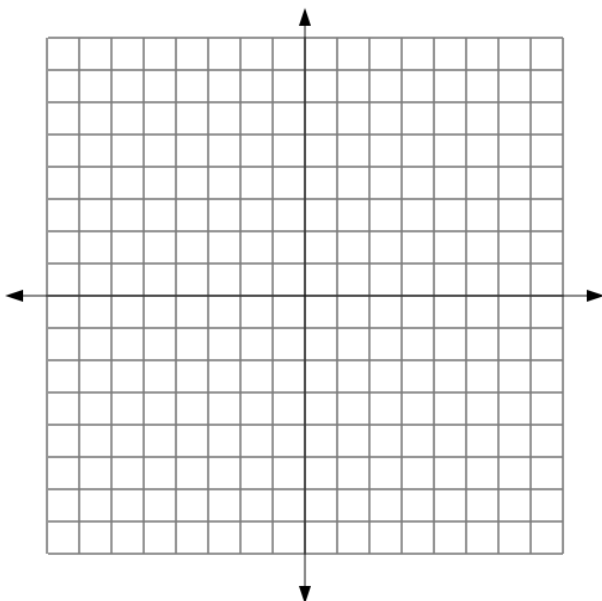
c) $x - 4y + 12 = 0$

d) $3x + 2y - 5 = 0$

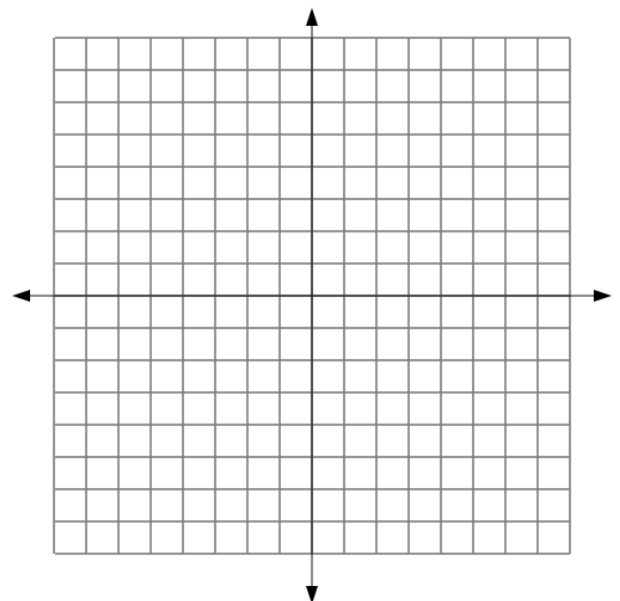
2. For each linear relation in question 1,

- Identify the slope and the y-intercept
- Use this information to graph the line

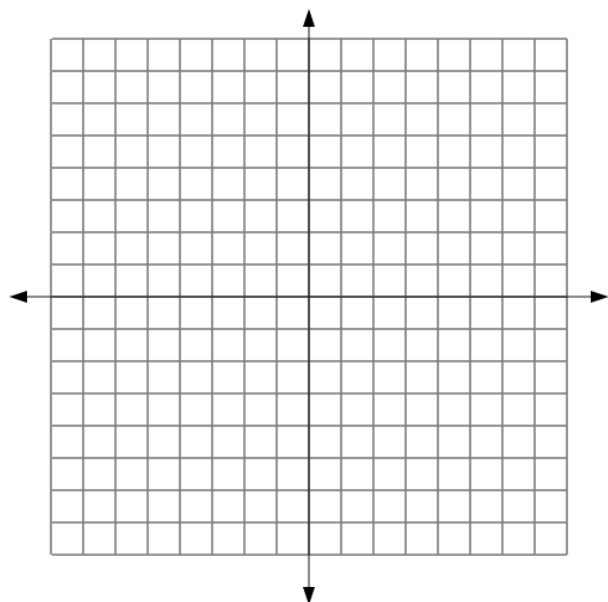
a)



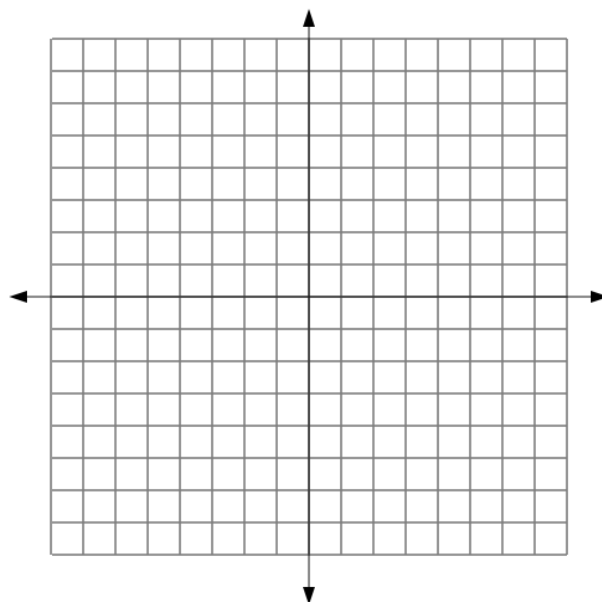
b)



c)



d)



3. Identify the slope and the y-intercept of each line.

a) $x + 3y - 3 = 0$

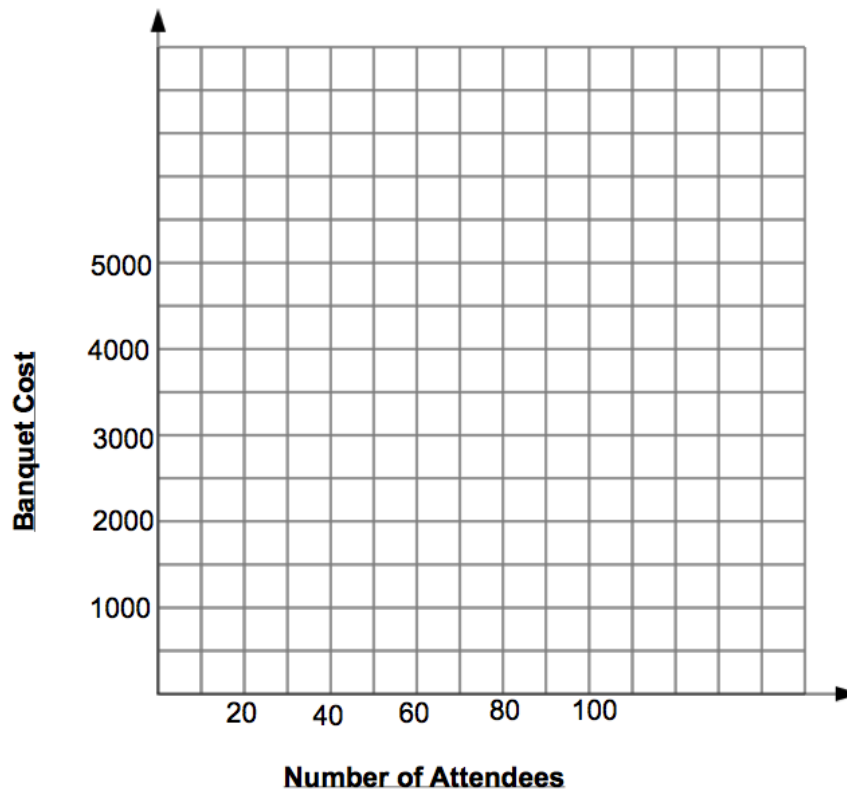
b) $2x - 5y + 8 = 0$

4. The Easy Event banquet hall charges according to the equation $40n - C + 250 = 0$

a) Express the equation in slope y-intercept form: $C = mn + b$

b) Identify the fixed and variable costs.

c) Illustrate the relation graphically



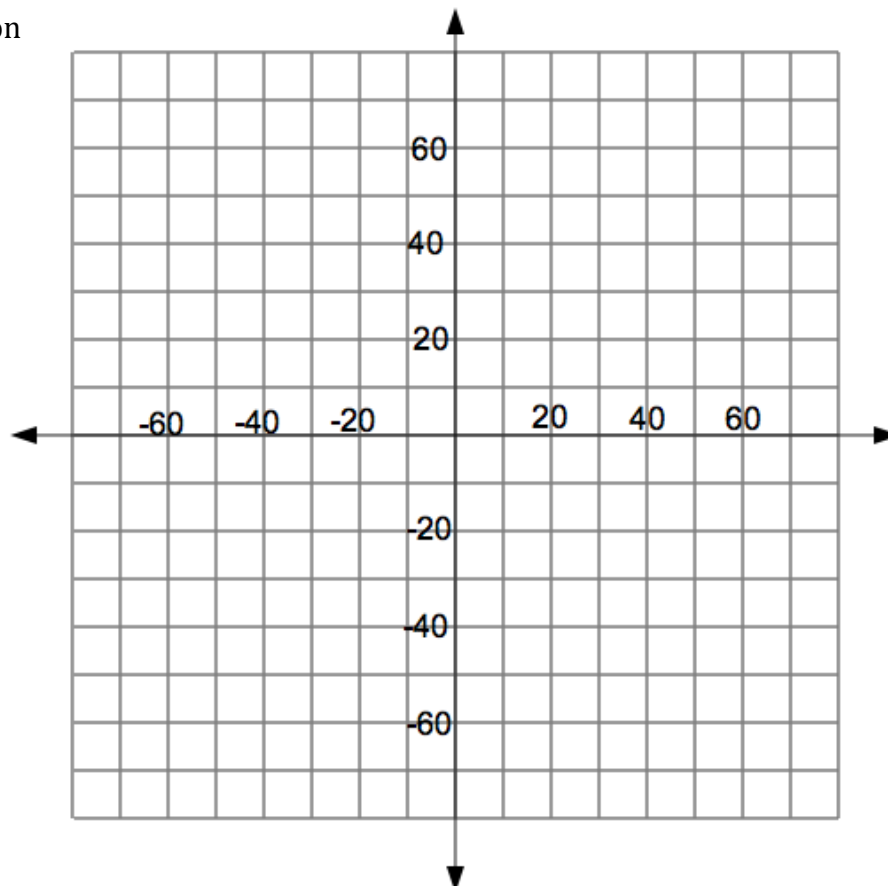
d) What is the rental cost if 100 people attend a soccer banquet?

5. There are a lot of factories in the Jean's home city. The equation $n - E + 15 = 0$ describes how much a worker might earn, E , in dollars per hour, according to the number of years experience, n . Find the hourly earnings of a beginning factory worker, and of a worker with 5 years experience.

6. The equation $9C - 5F + 160 = 0$ describes the relationship between temperature, C , in degrees Celsius, and temperature, F , in degrees Fahrenheit.

a) Express this equation in the form $C = mf + b$

b) Graph this relation



c) Identify the slope and the C-intercept and explain what they mean.

7. The Knights Banquet Hall uses the equation $25n - C + 1250 = 0$ to determine the cost for a hall rental. The Legions Banquet Hall uses the equation $30n + 995 - C = 0$ to determine the cost for their hall rental. In each case, C represents the cost, in dollars, which depends on n , the number of people attending.

a) Express each equation in slope y-intercept form: $C = mn + b$

b) Identify the fixed and variable costs for each hall.

c) What is the cost at each hall for a graduation banquet for 45 people?

8. [BONUS] You can change an equation from slope y-intercept form to standard form by rearranging terms in the equation. Rearrange each of the following equation into standard form. $Ax + By + c = 0$, and identify the coefficients A, B, and C. Hint: In standard form, the coefficient of the x-term is always positive and there are no fractions.

a) $y = -2x + 7$

b) $y = \frac{3}{4}x - 2$